Gaze Patterns in Individuals with Autism or Social Anxiety

Lucija Sorić[1], Anthony P. Atkinson[2], Anouschka van Dijk[1], Mariska E. Kret[1,3]
[1] Cognitive Psychology Unit, Leiden University; [2] Durham University; [3] Leiden Institute for Brain and Cognition

Background

- Emotion perception and recognition are **crucial for social functioning**.
- Faces represent an important source of information about others' emotion.
- **Autism spectrum** (ASD) and **social anxiety disorder** (SAD) are associated with impaired emotion perception.
- Research has mainly used region-of-interest analyses and static images, not **dynamic images** (videos).

Where do individuals with **ASD** and **SAD** look when viewing dynamic emotional faces? Does this differ from controls?

Methods

***N* = 106** (**control** = 40, **ASD** = 40, **SAD** = 26)
8 actors x 5 basic emotions, 40 videos

"What type of **expression** was exhibited by the person in the video?"

Preliminary Results

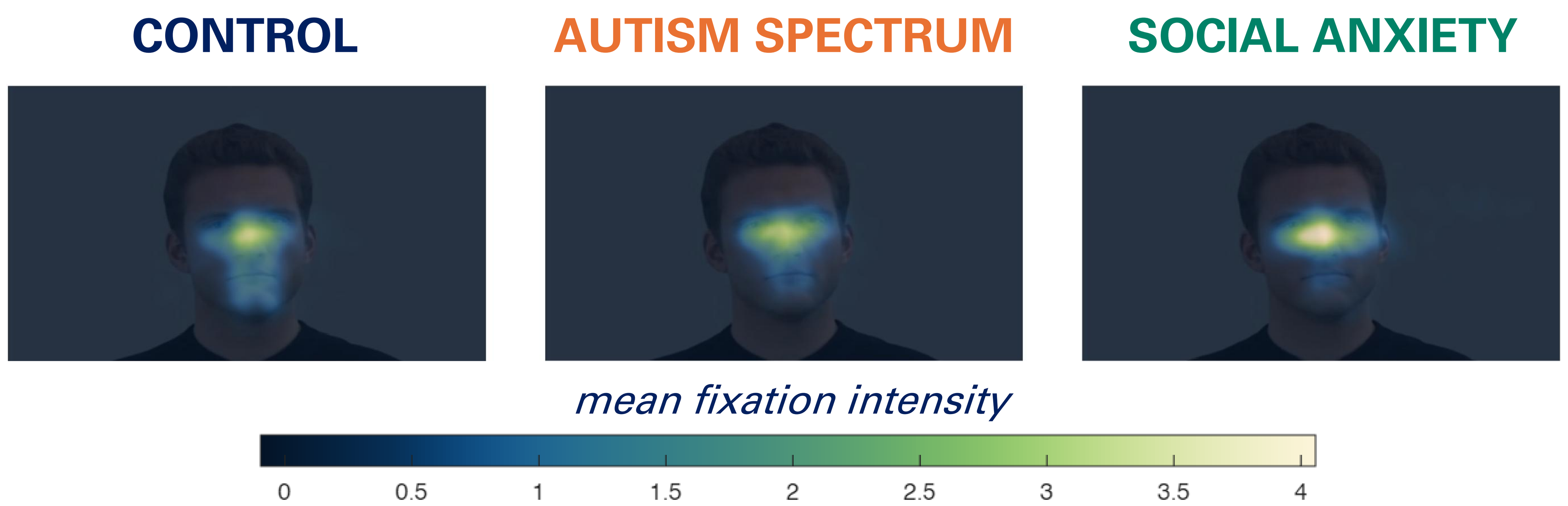


Figure 1

Happiness

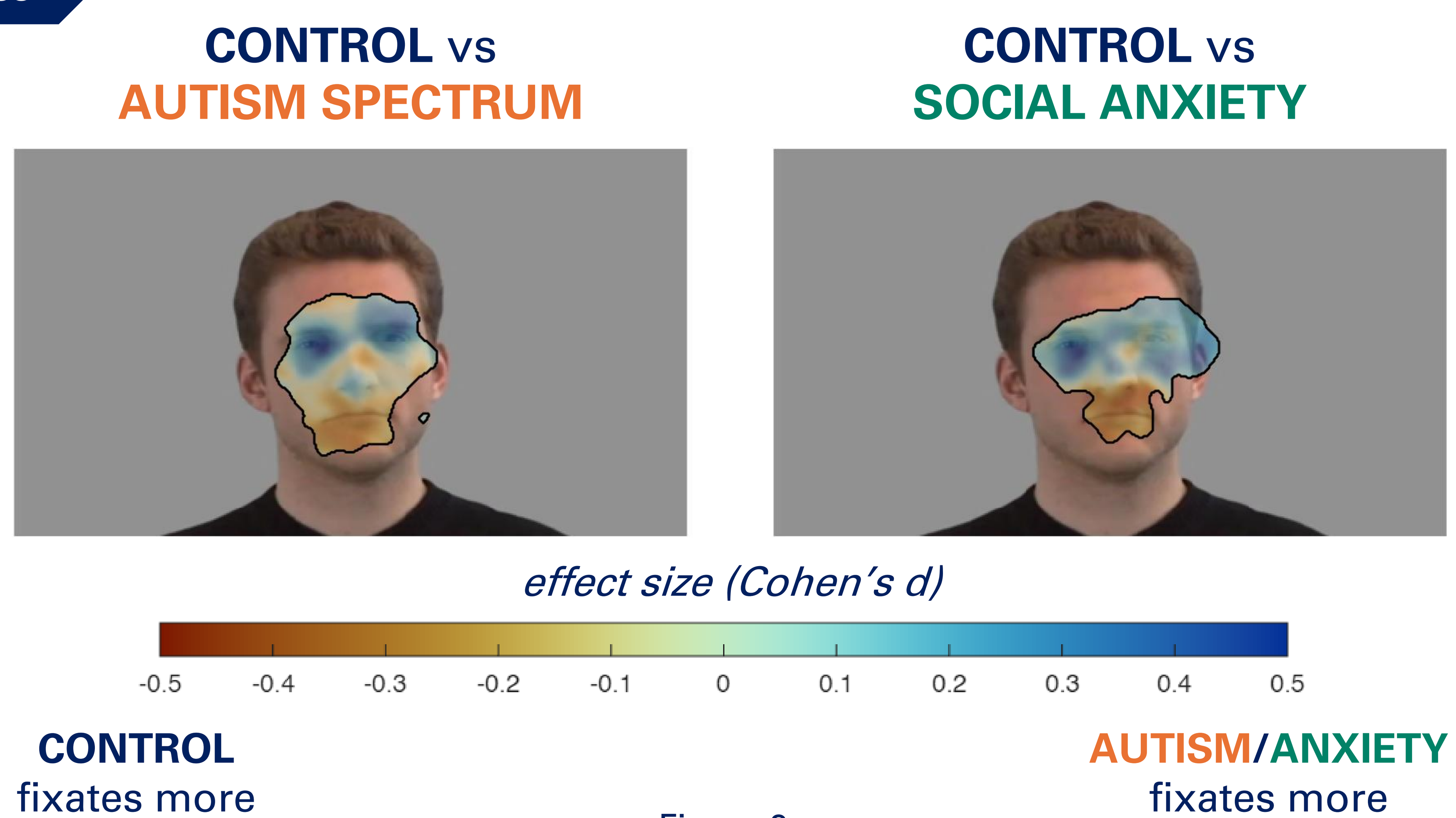
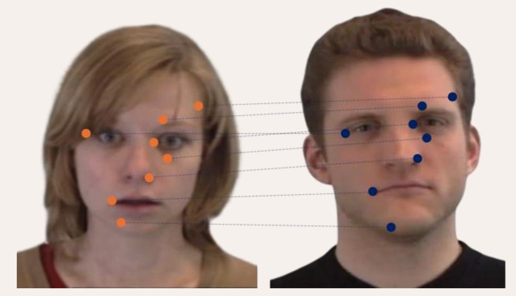


Figure 2

- Across all emotions (fear, sadness, happiness, anger, neutral) **ASD**/**SAD** groups show altered fixation patterns compared to **controls**, with small-to-medium ES (Figure 2).
- **ASD**/**SAD** groups preferentially fixate on the eye and nose regions (Figure 1, 2).
- **No significant differences** found between **ASD** and **SAD** groups across any emotion.

Analytical Approach



Normalizing, scaling and aligning fixations across actors

Creating fixation heatmaps with Gaussian smoothing

Linear models per pixel, contrasts between groups

Conclusion

- Across all emotions, **controls** disperse fixations more than **ASD**/**SAD** groups.

When looking at emotional faces, people on the **autism spectrum** and people with **social anxiety** fixate more on eye regions than **controls** do.

Scan for references & code

contact: **soricxlucija@gmail.com**